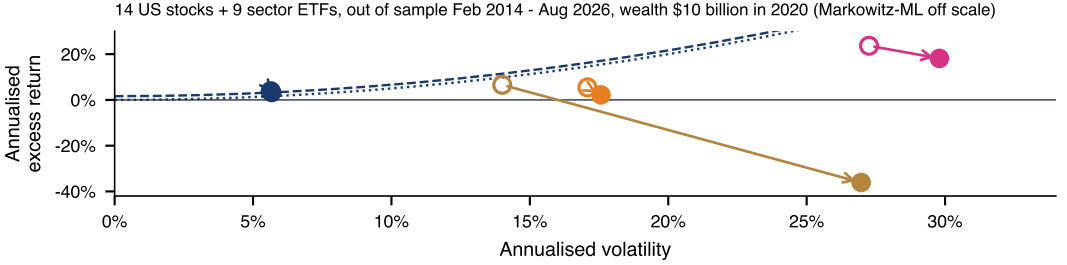
14 US stocks + 9 sector ETFs, out of sample Feb 2014 - Aug 2026, wealth $10 billion in 2020 (Markowitz-ML off scale)
Annualised excess return
20%
0%
-20%
-40%
0%
5%
10%
15%
20%
25%
30%
Annualised volatility
Utility = 0.016 (γ = 10)
Utility = 0.000 (γ = 10)
Portfolio-ML: net Sharpe 0.57, utility 0.016
Static-ML: net Sharpe 0.13, utility -0.132
Minimum variance: net Sharpe -1.34, utility -0.725
1/N: net Sharpe 0.61, utility -0.262
Gross of costs (hollow)